



Stationary Processes

Agenda

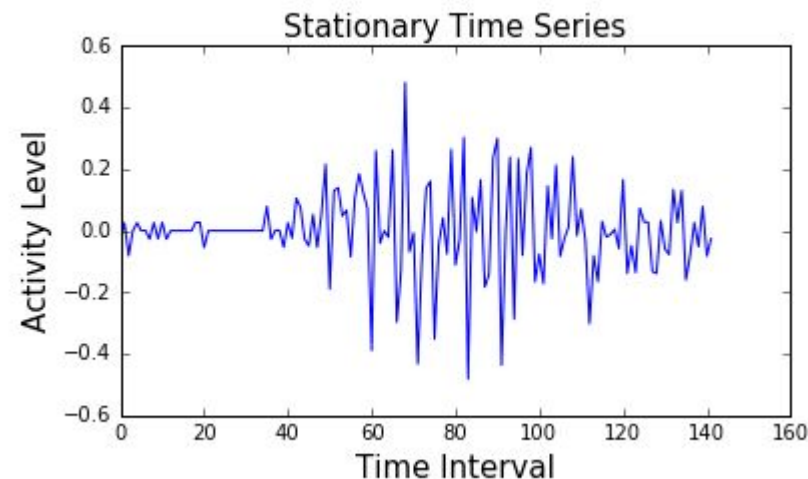
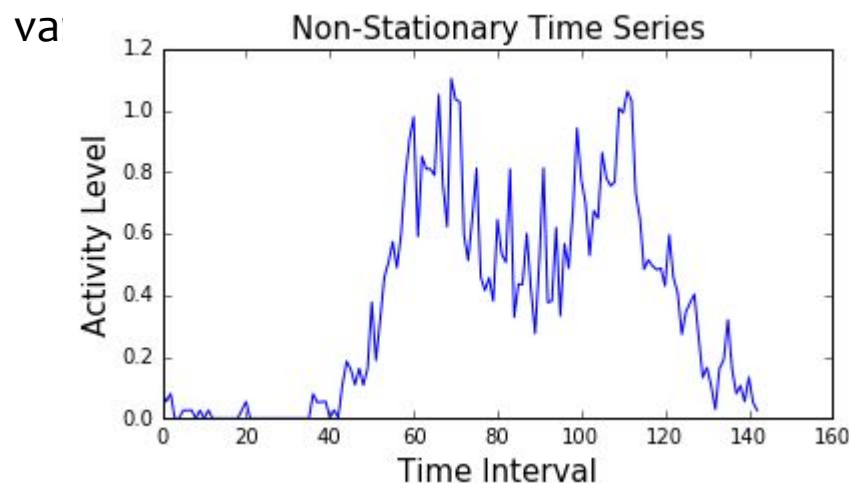
- Stationarity, Lags, and Introducing ARMA Models
- Differencing and Mastering ACF/PACF for Model Identification
- Hands-on: Core ARMA Workflow in Python



Stationarity, Lags, and Introducing ARMA Models

What is a Stationary Time Series?

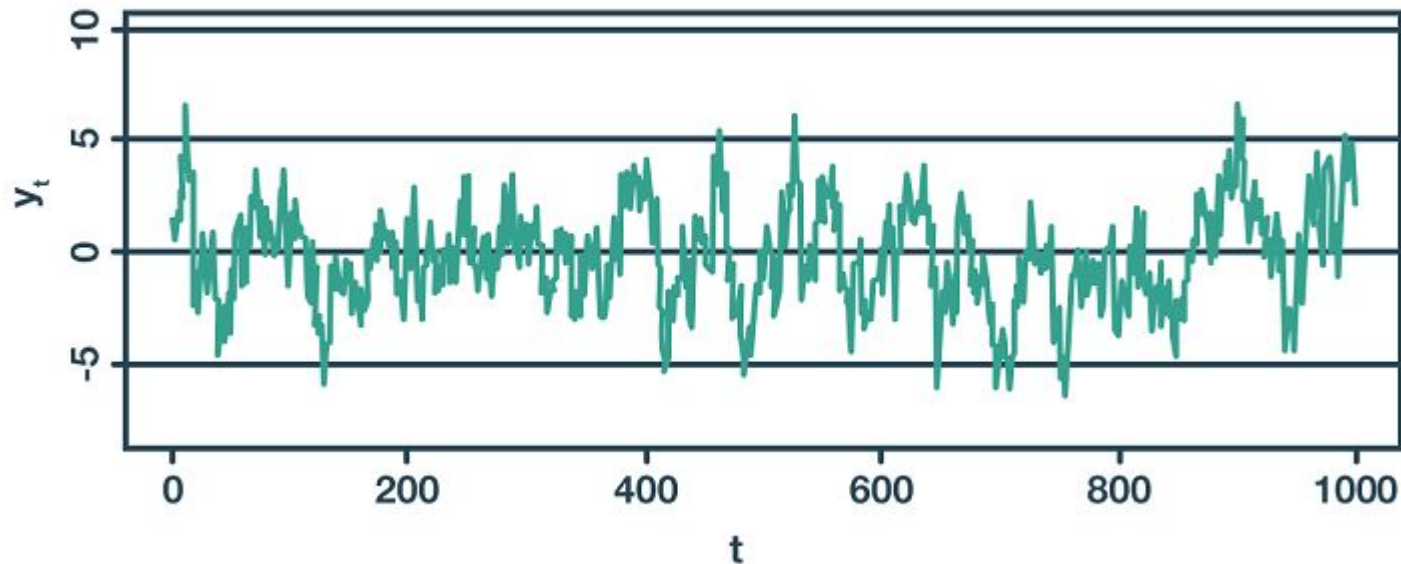
- A stationary time series is one whose statistical properties—mean, variance, and autocorrelation—are all constant over time.
- This means the series looks and behaves the same way regardless of the time period you observe it in.
- stationary series has no upward or downward trend, and its cyclical or seasonal patterns don't get bigger or smaller over time.
- Non-stationary examples: A series with a clear upward trend (e.g., a series with increasing variance over time).



Why is Stationarity So Crucial?

- Most of the statistical models we use for time series, including ARMA, are based on the assumption that the data is stationary.
- If a time series is not stationary, the predictions we make using models trained on it will be unreliable. We are essentially trying to predict future behavior based on past patterns that no longer exist or are changing.
- Stationarity allows forecasting to be valid.

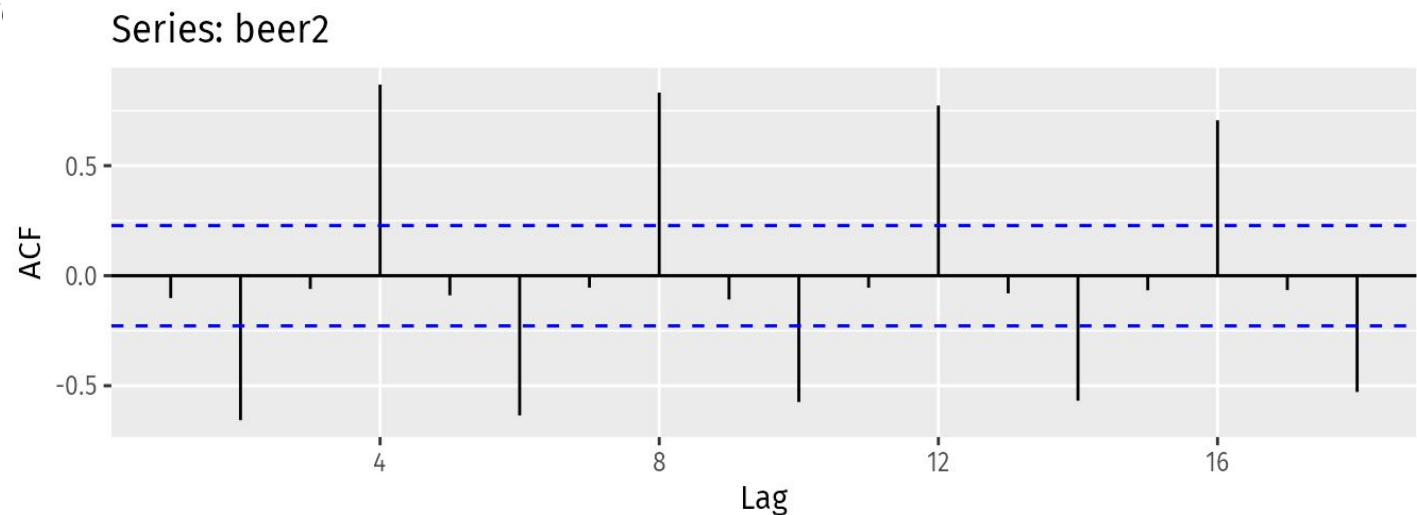
Stationary Time Series



Concept of Lag and Autocorrelation

- A lag is a specific past time point relative to the current observation. A lag of 1 refers to the value from the previous time period. A lag of 12 refers to the value from 12 time periods ago.
- Autocorrelation is the correlation of a series with a lagged version of itself. It measures how much the value at a certain time is related to the value at a past time.
- An Autocorrelation Function (ACF) plot shows the autocorrelation for different lags. In a stationary series, autocorrelation should converge to near zero as the lag increases.

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2},$$



How to identify a series is stationary

- Method 1: Visual Inspection: First, we can plot the series and look for obvious signs of non-stationarity, such as a trend, changing variance, or strong seasonality.
- Method 2: Formal Tests: To be sure, we use statistical tests like the Augmented Dickey-Fuller (ADF) test.
 - The null hypothesis (H_0) of the ADF test is that the series is non-stationary.
 - If the p-value is below a certain threshold (e.g., 0.05), we reject the null hypothesis and conclude the series is stationary.
 - We can also use the KPSS (Kwiatkowski-Phillips-Schmidt-Shin) test, which has the opposite null hypothesis.
 - The null hypothesis (H_0) of the KPSS test is that the series is stationary.
 - If the p-value is below a certain threshold (e.g., 0.05), we reject the null hypothesis and conclude the series is non-stationary.

Differencing

- Differencing is a simple but powerful technique to transform a non-stationary time series into a stationary one. You subtract the value at a previous time step from the current value.
- Differencing helps to remove trends and seasonality from the data. The "I" (Integrated) component of an ARIMA model represents the number of times you have to difference the series to make it stationary.
- Example:
 - A first-order difference is $Y_t' = Y_t - Y_{t-1}$.
 - This process effectively removes trends from the series. If seasonality is present, we might need a seasonal difference (e.g., $Y_t'' = Y_t - Y_{t-12}$ for monthly data).
 - We can apply differencing multiple times until the series becomes stationary.
- By focusing on the change from one period to the next, we remove the long-term upward or downward movement of the series, leaving only the stationary fluctuations.

ACF and PACF

- The ACF (Autocorrelation Function)
 - Shows the correlation between a series and its lagged versions. It tells us the total amount of autocorrelation at each lag.
 - Shows the total correlation between a time series and its lagged versions. It's like asking, "How much does today's value depend on the value from 1 day ago, 2 days ago, 3 days ago, and so on?" This correlation is both direct and indirect
- The PACF (Partial Autocorrelation Function)
 - Shows the correlation between a series and its lagged versions, after removing the effects of all the intermediate lags. It tells us the direct correlation between an observation and its past values.
 - Shows the direct correlation between a time series and its lagged versions. It answers the question, "How much does today's value depend directly on the value from 3 days ago, after accounting for the influence of the values from 1 and 2 days ago?"
- The Key: The patterns in these plots for a stationary series are like a fingerprint for the underlying AR and MA components.

Limitations of Exponential Smoothing

- Exponential smoothing models (like Simple, Holt, and Holt-Winters) are excellent for smoothing out random noise and capturing trends and seasonality.
- However, they are limited because they do not explicitly model the dependence structure between observations.
- Instead of using past values to predict the next value directly, they use an exponentially weighted average of past data. This can be less effective at capturing complex, long-term relationships within a time series.

Introducing ARIMA(p, d, q) Models

- ARIMA stands for Autoregressive Integrated Moving-Average.
- It's a powerful model that combines three different components to capture the data's internal dependencies.
 - The Autoregressive (AR) part, denoted as AR(p):
 - p is the order, or number of lags, to include in the model.
 - This component models the relationship between an observation and its past values. It says that the current value is a linear combination of its previous p values.
- The Integrated (I) part, denoted as I(d)
 - d is the order of differencing required to make the series stationary.
 - This component directly links the model to the process of differencing we discussed earlier.
- The Moving-Average (MA) part, denoted as MA(q)
 - q is the order, or number of lags, for the error terms.

Interpretation Rules for Model Identification

- AR(p) Process: A time series with only an Autoregressive component.
 - ACF: Dies down exponentially (the spikes gradually get smaller and smaller).
 - PACF: Cuts off sharply after lag p (there are significant spikes up to lag p, and then they abruptly drop to zero).
- MA(q) Process: A time series with only a Moving-Average component.
 - ACF: Cuts off sharply after lag q.
 - PACF: Dies down exponentially.
- ARMA(p, q) Process: A time series with both AR and MA components.
- ACF and PACF: Both die down exponentially without a clear cutoff. This indicates a mix of AR and MA behavior.

White Noise

- What is White Noise? A white noise series is one where the data points are independent and identically distributed. There is no pattern or correlation in the data.
- The Plots:
 - ACF and PACF: Both plots will show no significant spikes at any lag. All the correlations will be very close to zero, staying within the confidence intervals.
- Why it Matters: A series that has been successfully differenced and modeled should have residuals that resemble white noise. This is how we know our model has captured all of the underlying structure in the data.

Autoregressive models

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t,$$

- In an autoregression model, we forecast the variable of interest using a linear combination of *past values of the variable*.
- The term *autoregression* indicates that it is a regression of the variable against itself.
- For an AR(1) model:
 - when $\phi_1=0$, y_t is equivalent to white noise;
 - when $\phi_1=1$ and $c=0$, y_t is equivalent to a random walk;
 - when $\phi_1=1$ and $c \neq 0$, y_t is equivalent to a random walk with drift;
 - when $\phi_1 < 0$, y_t tends to oscillate around the mean.

Moving average models

- Rather than using past values of the forecast variable in a regression, a moving average model uses past forecast errors in a regression-like model.

$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \cdots + \theta_q \varepsilon_{t-q}$$